Notes on the v11 TWS forecast

Zindi / ITU challenge: A Step Ahead of Drought.

This is the write-up that goes with `TWS_Solution_v11.ipynb`. I am answering the six points in the Trustworthiness Evaluation document. Numbers below come from the v11 run in `outputs_v11/`, not from memory.

What the model does: predict next month's GRACE total water storage anomaly on the 1° land grid. In the test file TWS is missing a lot of the time, so many rows are really 3–7 months ahead, not one.

Rule I followed: at month t, nothing later than t goes into the model. The flag `USE_TARGET_MONTH_FORCING` is False.

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1. Data and model bias

GRACE itself is biased toward large scales. The 1° grid is an interpolation of a spherical-harmonic field, so neighbouring cells are not independent. I did not treat them as if they were.

The bigger problem is time. In training, missing TWS usually lasts one or two months. In test, the gaps are 3, 4, 5, even 18 months. About 28% of test rows sit at horizons the training panel never really saw. v8 tried to copy that gap pattern in validation. It made the local numbers look honest and then scored worse on the leaderboard (0.657 vs 0.653). So I stopped using that trick.

The SPEI and soil-moisture columns in Test.csv only exist for months that are in the file. Across a long void they are empty. GDO SPI and fAPAR are complete for every month, which is the main reason they are in v11.

Geography, measured on the holdout (prediction minus truth; negative means too dry):

Tropics: bias −0.095, RMSE 0.696 (n = 110,613)

Southern extratropics: −0.104, RMSE 0.649

Northern mid-latitudes: +0.009, RMSE 0.607

High northern latitudes: +0.049, RMSE 0.573

By |latitude|: 0–15° RMSE 0.655; 15–30° 0.725 (worst); 45–90° around 0.56–0.58. The model is too dry in the tropics. Persistence is also worst there, and we still beat it in every band (gain 0.21–0.25 RMSE).

I tried a per-band additive correction. It helped the calibration window and failed on holdout. v8's per-horizon calibration cost +0.004 on the public board. Both corrections are off. I would rather show the tropical bias than hide it with a patch that does not travel.

No lat/lon features, no per-cell mean. Cells the model has never seen still get the same physical descriptors (snow fraction, frozen soil, SPI, ERA5 storage). Recency weights (tau = 120 months) pull the fit toward the later, noisier GRACE era, which is closer to test.

I did not use SEAS5. A side test showed it made RMSE worse.

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2. Model transparency

This is not a neural net. There is a two-timescale Kalman filter (slow memory $\varphi \approx 0.92$, fast ≈ 0.52) that I can write down, and then trees that correct it.

Four models in the stack, weights fitted on CAL only:

- XGBoost residual 0.36
- LightGBM residual 0.23
- LightGBM direct 0.22
- CatBoost residual 0.19

Plus a LightGBM specialist on 1-month-ahead rows.

TreeSHAP on 20k holdout rows. Top features:

1. e_S_s40 (0.124) — ERA5-Land column storage, smoothed. This beat the Kalman base.
2. base (0.094) — the Kalman forecast itself
3. g_spi3_s40 (0.077) — GDO SPI-3, regional
4. clim_delta (0.074) — seasonal step
5. e_S_s25 (0.037) — same storage, tighter smooth

ERA5 LAI is almost unused (it is a climatology, not a real vegetation series). That is why I added MODIS fAPAR from GDO. Snow and deep soil show up in SHAP because SPEI and near-surface soil moisture cannot see them.

ERA5 features are 23% of total |SHAP|, GDO 14.5%. Full table: `outputs_v11/feature_importance.csv`. Plot: `shap_summary.png`.

A hydrologist can look at k_pred / k_level on a cell and see what the filter believed before the trees moved it.

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3. Approach reusability

To score a new cell you need: TWS history on that cell, ERA5-Land, and GDO SPI/fAPAR. You do not need to retrain a latitude dummy. That was deliberate.

All extra data is Copernicus and public. CDS for ERA5-Land, GDO over plain HTTP. No paid API. Scripts: `gdo_download.py`, `gdo_prepare.py`, `gdo_prepare_spi.py`, `era5_extra_download.py`.

Kalman and EOF only look backward. A new month can be appended without rewriting the past.

Flags if someone wants a cheaper run: `TWS_FAST=1` (one seed, LightGBM only), `TWS_SMOKE=1` (tiny structural test). `USE_ERA5` and `USE_GDO` can be switched off.

This is built for monthly TWS on a regular grid. I would not drop it onto a river basin at 1 km and expect it to work. The useful bits to copy are the causal filter, residual trees, physical features instead of coordinates, and not trusting a bias correction that only works on the window you tuned.

Packages are ordinary open source (LightGBM, XGBoost, CatBoost). Versions in `requirements.txt`.

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4. Sustainability and efficiency

One laptop, no GPU.

- CPU: Intel i7-1185G7, 6 threads used
- RAM: ~23 GB of 32 GB
- Wall clock: 40,341 s (about 11.2 hours)
- CodeCarbon: 56.5 g CO2e, 0.33 kWh
- OS: Windows 11, Python 3.12.10

Almost all of that time is training. Writing the 280,961 test rows takes a few minutes.

I did not use a deep model. fAPAR arrives as 2.8 GB and is averaged down to a 34 MB 1° cache before training. Emissions file: `outputs_v11/emissions.csv`.

This is heavier than a single LightGBM. The extra trees are worth about 0.001–0.002 RMSE. If an agency needed a daily refresh they would run FAST mode or skip XGBoost/CatBoost. For a monthly GRACE gap-fill, 11 hours on a CPU once a month is acceptable.

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5. Innovation

Most of the movement on this task came from information, not from a fancier learner.

What I actually checked:

v8 copied the test gap structure into validation. Leaderboard got worse (0.657123). Calibration was the damage (+0.004). The other structural changes netted about zero.

v9 kept the v7 base, fixed one bug, added recency weights, forced calibration off. Public LB 0.651802 (best of the scored versions). The bug: EOF deltas were taken against month anc−1, which is missing at test time, so those features were filled in training and NaN at prediction. Differencing against the last reconstructed month fixed it. The dense holdout could not see that bug. Local validation lied in both directions on this problem.

v11 is v9 plus GDO. A small LightGBM test of every extra source on disk (`diagnostics/07_value_of_information.py`):

- SEAS5 forecast of the target month: +0.0017 (worse)
- SPEI/SM from the t+1 row: −0.0004 (noise)
- GDO SPI-3/6 + fAPAR: −0.0032 (only real gain)
- Perfect target-month ERA5-Land: +0.0008

So the leftover error is not “missing weather.” I did not add forecasts. SPI from GPCC, CPC and JRA-3Q only agree at r ≈ 0.53–0.64. I use the mean and keep the spread as a “we don’t know precip here” feature.

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6. Practicality and robustness

Causal on purpose. GDO SPI for month t is published in the first days of t+1, same as the organisers’ SPEI_t. ERA5 features stop at t. No GRACE in those products.

Holdout RMSE by horizon stays in a narrow band: h=1 0.615, h=2 0.625, h=3 0.624, h=4 0.639, h=5 0.617, h=6 0.647, h=7 0.602. Gain vs persistence is larger at long horizon (0.17 at h=1, 0.33 at h=7), which is what you want when GRACE has been dark for half a year.

Final holdout RMSE 0.6229 vs persistence 0.855. Climatology 1.113. Kalman alone 0.788. The trees do the rest.

Public scores I actually submitted along the way: v7 0.653156, v8 0.657123, v9 0.651802. v11 is this notebook. Local holdout is not the public LB — different era, different gap mix — so I am not pretending 0.6229 is a board score.

Known failure: tropical dry bias ~0.09. Do not slap a constant on the tropics. If this went operational I would publish the per-band table next to the map.

Runs on a normal laptop. Seeds are fixed (LightGBM 1,2,3; numpy 7/5/6/11). If ERA5 or GDO files are missing the notebook prints that and continues. If CatBoost fails to import, LightGBM still writes a submission.

Predictions are raw floats. I did not round them or set thresholds.

How to rerun it

Machine: local Windows laptop, specs above. Not Colab.

```
pip install -r requirements.txt
```

Put `Train.csv` and `Test.csv` next to the notebook. Need also:

- era5/era5land_YYYY.nc
- era5_extra/era5land_extra_YYYY.nc
- gdo/gdo_1deg_monthly.npz
- gdo/spi_multi_1deg_monthly.npz

Then run `TWS_Solution_v11.ipynb` top to bottom, or `python tws_v11.py`. Output: `outputs_v11/submission_v11.csv`.

Main LightGBM settings: lr 0.03, 255 leaves, min_data 100, feature_fraction 0.75, early stopping on CAL, final refit 689 rounds.

Ladder on the holdout (RMSE)

climatology 1.113

persistence 0.855

Kalman 0.788

LightGBM direct 0.639

LightGBM residual 0.638

XGBoost 0.637

CatBoost 0.635

stack 0.635

stack + smooth + h=1 specialist 0.623 ← what I ship

+ calibration 0.625 (worse; left off)

+ band de-bias 0.623 (left off)